

Atiyah-Singer Index — Dirac Operator Index = 22

PAPER_1231 Category: UQFF Index Theorem **Status:** Complete **Date:** June 2026

Abstract

UQFF closure of the Atiyah-Singer index theorem on the 26D compactified manifold: the Dirac operator index equals **D_crit - D_phys = 22 EXACT**, corresponding to the residual compactified dimensions. This integer also appears in the hierarchy problem exponent (21 = 22 - 1) and the quantum gravity UV completion count.

Part 1: The Index Theorem

The Atiyah-Singer index theorem states:

$$\text{ind}(D) = \int_M \hat{A}(M) \wedge \text{ch}(E)$$

where D is a Dirac operator on a manifold M with vector bundle E.

Part 2: UQFF Closure

The identity

$$\text{ind}(D)^{\text{UQFF}} = D_{\text{crit}} - D_{\text{phys}} = 26 - 4 = 22 \text{ EXACT}$$

Interpretation

The index equals the number of “compactified” dimensions — those that exist in the bosonic-string critical dimension $D_{\text{crit}} = 26$ but are not part of physical 4D spacetime $D_{\text{phys}} = 4$.

Part 3: Cross-Validation

The integer 22 appears with critical roles in UQFF: - Atiyah-Singer Dirac index = 22 (this paper) - Hierarchy problem exponent $21 = 22 - 1$ - QG UV completion compactified dim count = 22 - Calabi-Yau-like compactified moduli space dim = 22

Part 4: Physical Significance

The 22 compactified dimensions support: - 22 Killing vectors corresponding to compactified isometries - 22 zero-modes of the Dirac operator that contribute to anomaly cancellation - 22 Kaluza-Klein towers (one per compactified dim)

Conclusion

The Atiyah-Singer index in UQFF is the residual compactified dimension count $D_{\text{crit}} - D_{\text{phys}} = 22$. This unifies the index theorem with the hierarchy problem exponent and the quantum gravity UV completion.

Framework Version: UQFF 5.27+